sample.bib

SPENSER MILLBURN

Embedded Software Engineer / Roboticist

spenser.engineer
Spenser Millburn
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505-604-5705
Boston, MA

Dedicated, enthusiastic and ambitious professional with excellent comprehension and retention ability. Breadth and depth knowledge in a broad spectrum of technical fields concentrated within Robotics, EE, CS and Mechanical Design.

EXPERIENCE

Embedded Software Engineer
Alert Innovation > Walmart Global Tech

September 2022 – Current | Boston, MA

- Designed, implemented, and unit tested performant hierarchical state machines with modern C++17 in a real time QNX microkernel environment.
- Developed device drivers for critical I2C and SPI based sensors, enabling accurate localization of autonomous robots in production (50k units, 99.7% uptime).
- Spearheaded a company-wide transition towards Azure IoT based configuration management. Owned SDK integration to codebase development for embedded (C++), backend (Python), and frontend (React)
- Collaborated cross-functionally to establish data pipelines between IoT Hub and robot software, enabling efficient data exchange and real-time cloud analytics.
- Dockerized the embedded development environment, resulting in streamlined workflows, centralized tooling, and accelerated development cycles.

Robotics Engineer
Building Machines

March 2022 – September 2022 | Boston, MA

- Full ownership and system design of a high powered (480V 50Hp) robotic fluid delivery module for additive manufacturing.
- Designed implemented, tested and integrated a full stack React GUI to override physical control panel.
- WebSocket interface and Flask python backend. Systemd linux service management.
- IOT integration of multiple I2C and SPI sensors/actuators with custom PCBs and C++ device drivers. Modalities: Laser Time-Of-Flight, Depth CV, Pressure Transducers, Solenoid Valves, BLDC motor, relays
- Data acquisition,analysis and visualization with JupyterLab and Python.
- Electromechanical design, build, test of fixtures

Engineering Technician > Test Engineer
Owl Labs Inc

March 2021– May 2022 | Boston, MA

- Rapid prototyping, prototype assembly and design. Supported 30+ FTEs.
- Professional fixture design, fabrication, testing and deployment. Oversaw CM production line integration.
- Thermal data acquisition and Test Engineering.
- Customer facing Embedded Linux software development support.
- Data acquisition, analysis and visualization not limited to thermals, acoustics and airflow.

Robotic Assembly Technician
Sartorius Stedim Biotechnology

August 2018 – March 2021 | Albuquerque, NM

EDUCATION

B.S. in Mechatronics Engineering
Northeastern University

Sept 2020-23 | **3.9**

COMPUTER SCIENCE

Control Systems | Simulation | opencv | SLAM | Kinematics

C++ 21 | C | Rust | OOP | CMake | Eigen+LA | Python | Numpy | Pandas

Linux + Kernel | Docker | Kubernetes | Azure | AWS | Terraform

TS | React.js | Node.js | MongoDB | FireBase | SQL | Flask | bash

ELECTRICAL

Advanced SMT Soldering | Altium Designer | I2C | SPI | CAN | Arduino/RPi | FPGA | Oscilloscope | Logic Analyzer | Bed of Nails

MECHANICAL

OnShape | Creo | Solidworks | AutoCAD | CNC/CAM Machining (Mill/Lathe) | DFM | Fluidics/Microfluidics | Thermal Analysis

PROJECTS

Husky Quadruped
Northeastern Robotics, PM & Software Lead

- Project Management of 75+ student engineering organization, development of advanced quadruped research platform.
- Significant technical contributions to CAPSTAN servo mechanisms, ROS infrastructure, SLAM, Computer Vision, Inverse kinematics.